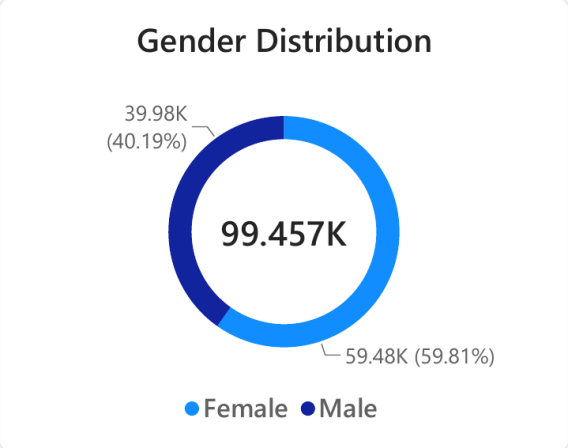
PRDA-04

Customer Data Analysis Report

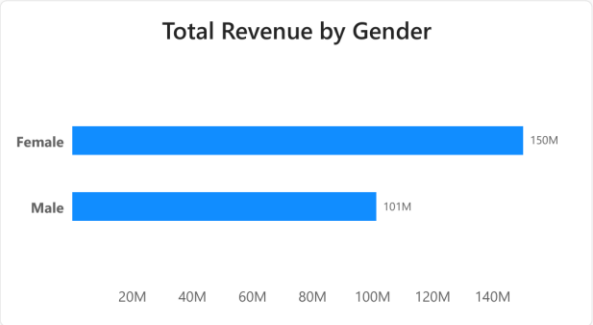
Total Revenue	**251,505,794**
Total Units Sold	**298,712**
Total Transaction	**99,457**
Average Basket Value	**2,529**

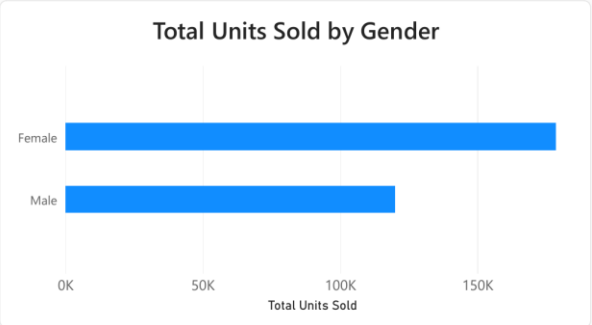
Gender Distribution



The gender distribution shows that female customers accounted for **59,482 transactions (59.81%)**, while male customers accounted for **39,975 transactions (40.19%)**. Female customers generated approximately **150.2 million TL** in revenue, compared with **101.3 million TL** generated by male customers. Overall, female customers had a higher share of both transactions and revenue.

Revenue and Units Sold by Gender





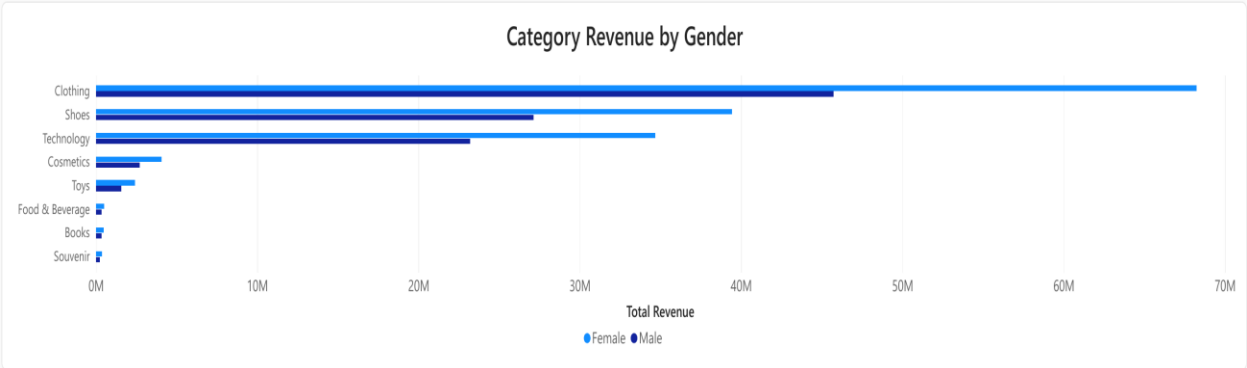
Which gender did we sell more products to?

Female customers purchased **178,659 units (59.8%)**, compared with **120,053 units (40.2%)** purchased by male customers.

Which gender generated more revenue?

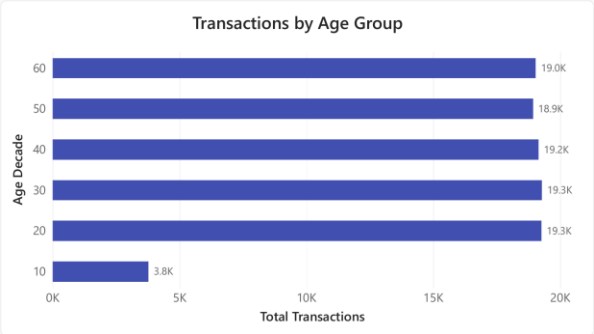
Female customers generated higher revenue of **TL 150.2M (59.7%)**, compared with **TL 101.3M (40.3%)** from male customers. Therefore, females contributed more to both units sold and total revenue.

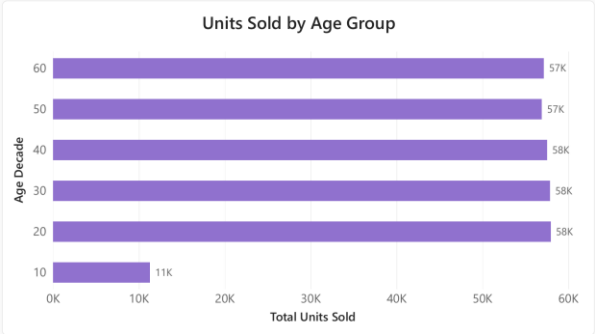
Category Revenue by Gender



Clothing generated the highest revenue for both females (**TL 68M**) and males (**TL 46M**), followed by Shoes (**TL 39M** female; **TL 27M** male) and Technology (**TL 35M** female; **TL 23M** male). Female customers generated higher revenue than male customers across all major categories, with the largest contribution coming from Clothing. Overall, Clothing, Shoes, and Technology are the leading revenue-generating categories for both genders.

Shopping Distribution by Age

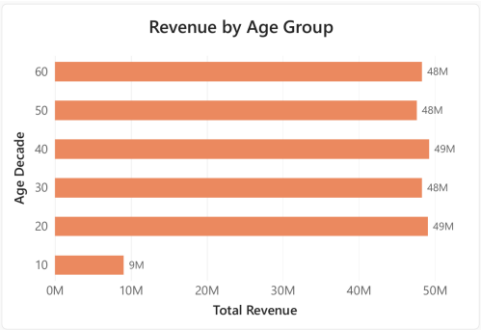


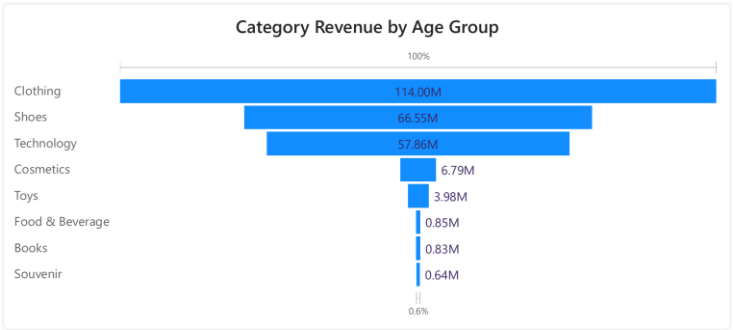


Which age group did we sell more products to?

Customers aged **20–69** are almost level at **18.9K–19.3K** transactions and **57K–58K** units per decade. The 20s lead narrowly on units **(57,949).** The "10" group is small only because it covers ages **18–19**, two years instead of ten; per year of age it matches the rest (about **1,890** transactions per age year against **1,890–1,930**).

Revenue and Category Mix by Age

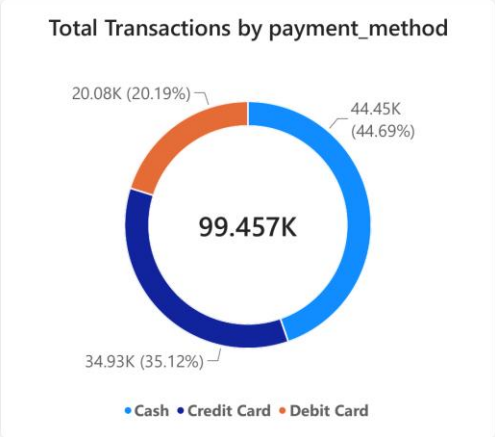


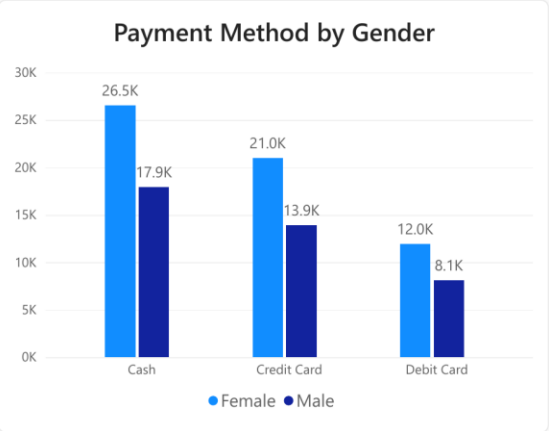


Which age group generated more revenue?

The 40s by a small margin **(TL 49.2M)**, then the 20s **(TL 49.0M**), with the 30s, 50s and 60s within **TL 47.6M–48.3M**. Clothing is **44–47%** of revenue in every group. Technology shifts with age: **19.4%** among ages **18–19**, **24.2%** in the 30s, and **22–23%** elsewhere.

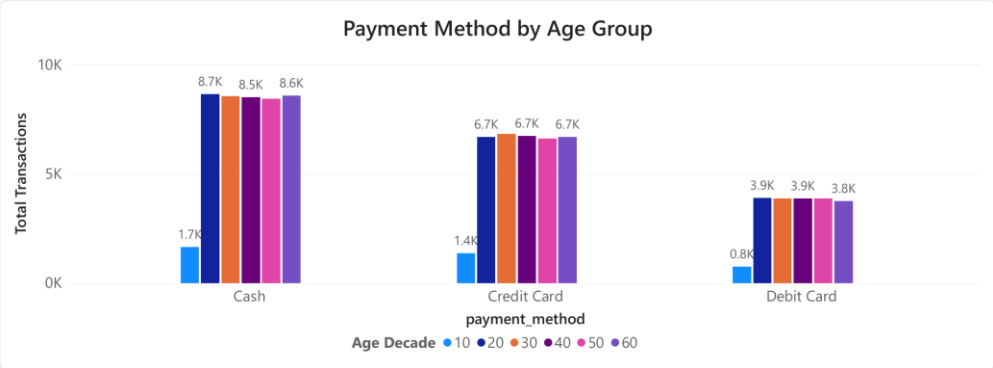
Payment Method Distribution

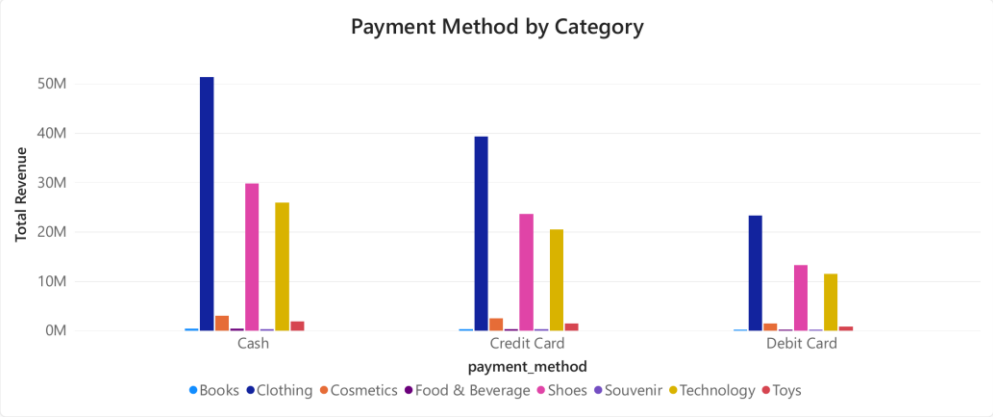




Cash is the most used method with **44,447** transactions **(44.69%)**, then credit card with **34,931 (35.12%)** and debit card with **20,079 (20.19%)**.

Payment Method by Age and Category





Does payment method relate to other columns? No meaningful relationship. The 45 / 35 / 20 split (cash / credit / debit) holds within 1–2 points across genders, every age group and all eight categories. Chi-square tests confirm no significant association with gender (p = 0.26), age group (p = 0.67) or category (p = 0.87). Category bars differ in height only because category sizes differ, not because of payment choice.

Insights & Recommendations

CUSTOMER OVERVIEW

- 99,457 transactions, 298,712 units and TL 251.51M revenue; average basket TL 2,529 (about 3 units per purchase).
- Revenue was flat between 2021 and 2022 (about TL 115M each year); 2023 data ends on 8 March.

GENDER

- Women make 59.8% of transactions and 59.7% of revenue; men 40.2% and 40.3%.
- Spend per transaction and category mix are the same for both genders, so the difference is visit frequency.

AGE

- Every group from 20 to 69 contributes roughly one-fifth of transactions and revenue; there is no dependence on one generation.
- Technology share is lowest among 18–19 year olds (19.4%) and peaks in the 30s (24.2%).

CATEGORIES

- Clothing, Shoes and Technology produce 94.8% of revenue from 49.8% of transactions.
- Cosmetics, Toys, Food & Beverage, Books and Souvenirs make 50.2% of transactions but only 5.2% of revenue, because unit prices are low (TL 16–122).

PAYMENT

- Cash is still the leading method (44.7%), and preferences are the same across all customer groups.

MALLS

- Mall of Istanbul (TL 50.9M) and Kanyon (TL 50.6M) each generate about 20% of revenue; the seven smallest malls generate TL 12–25M each. Basket size is similar everywhere (TL 2,485–2,579), so the difference is traffic.

RECOMMENDATIONS

- **Target marketing at women, but do not assume higher spend per visit.** They drive about 60% of traffic and revenue at every mall; for men, focus on bringing them into stores more often, since their basket is already equal.
- **Protect the three core categories.** Prioritise stock depth, promotions and mall placement for Clothing, Shoes and Technology, and watch the concentration risk (94.8% of revenue).
- **Use low-price categories as add-ons.** Cosmetics, Toys and Food & Beverage bring frequent visits; test bundles pairing them with Clothing or Shoes to lift basket value.
- **Grow Technology among under-30s.** It is 19–22% of revenue for ages 18–29 against 24% for the 30s; student or first-job offers and instalment plans could close the gap.
- **Treat payment method as an operations choice, not a segmentation tool.** Preferences are uniform, so card-linked cashback or instalment offers could reduce the 45% cash share if lower handling cost or better customer data is the goal.
- **Review the lower-traffic malls.** Seven malls earn TL 12–25M against about TL 50M for the top two at a similar basket size; examine footfall, store mix and local marketing.
- **Track the same views regularly.** Repeating this dashboard each quarter would catch shifts in gender, age or category mix early.

LIMITATIONS & SCOPE NOTES

- Revenue is quantity × unit price in TL. Discounts, returns and costs are not included, so figures are sales, not profit.
- Each invoice number and customer ID appears once, so repeat purchase, loyalty and customer lifetime value cannot be analysed.
- The “10” age decade covers only ages 18–19, and 2023 is a partial year (data ends 8 March). Gender is recorded as Female or Male only. The analysis is descriptive, not predictive.